

Value Iteration Problem

Title: Solving a Stochastic Markov Decision Process using Value Iteration

Problem Statement

Consider the following Markov Decision Process (MDP):

- **States:** $S = \{A, B, C, D, E, G\}$, where G is an absorbing goal state.
- **Actions:**
 - From A : $\{a_1, a_2\}$
 - From B : $\{a_1, a_2\}$
 - From C, D, E : $\{a_1\}$
 - From G : $\emptyset$
- **Transition Model:**
 - $T(A, a_1) : B \text{ (0.7), } C \text{ (0.3)}$
 - $T(A, a_2) : C \text{ (0.6), } D \text{ (0.4)}$
 - $T(B, a_1) : C \text{ (0.8), } E \text{ (0.2)}$
 - $T(B, a_2) : D \text{ (1.0)}$
 - $T(C, a_1) : E \text{ (0.9), } G \text{ (0.1)}$
 - $T(D, a_1) : E \text{ (0.7), } G \text{ (0.3)}$
 - $T(E, a_1) : G \text{ (1.0)}$
- **Cost Model:** Each transition incurs a cost of 2.
- **Discount Factor:** $\gamma = 0.95$

Tasks

1. Write the **Bellman optimality equation** for this MDP:

$$V^*(s) = \min_{a \in A(s)} \sum_{s'} T(s, a, s') [C(s, a, s') + \gamma V^*(s')]$$

2. Implement **Value Iteration** to compute the optimal value function $V^*(s)$.
3. Determine the optimal policy $\pi^*(s)$ for each state:

$$\pi^*(s) = \arg \min_{a \in A(s)} \sum_{s'} T(s, a, s') [C(s, a, s') + \gamma V^*(s')]$$

4. Report: the final value function, the optimal action per state, and the number of iterations.

Expected Outcome

You should obtain the optimal state values and the policy that minimizes the expected cost to reach the goal.